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Configurable audio and video sectors

Abstract

Systems and methods for configuring audio and video sectors of a camera. An audio analytics system includes a first computer configured to receive information from a camera and a microphone. The first computer includes an electronic processor. The electronic processor receives data including an image of a location and an audio of the location and generates a view of the location. The electronic processor sets, in response to receiving a first user input, a first sector of the view of the location, and sets, in response to receiving a second user input, a second sector of the view of the location. The electronic processor adjusts, in response to receiving a third user input, a configuration of the first sector, and adjusts, in response to receiving a fourth user input, a configuration of the second sector.

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Background/Summary

FIELD

(1) Embodiments, examples, and aspects relate to, among other things, systems and methods for configuring audio and video sectors of a camera.

SUMMARY

(2) Monitoring systems often use a combination of cameras and microphones to monitor a location. An operator of the monitoring system may want to monitor and configure multiple viewpoints of

the cameras independently, rather than setting a configuration of the entire viewpoint of each camera. For example, on a moving camera, the operator may configure one position or viewpoint to be sensitive to motion and mask off an area. Another position or viewpoint is configured to run an object detection algorithm. Yet another position or viewpoint is configured for low lighting scenarios. When an operator does not wish to capture conversations or sensitive audio in a given sector, audio recording may be disabled, or noise may be input into the audio for the specified sector

(3) Examples, instances, and aspects described herein provide, among other things, for a means to allow a user to monitor and configure viewpoints around a monitoring device independently. Both video and audio analytics are independently configured for each viewpoint or sector within the camera view.

(4) One example provides an audio analytics system having a first computer configured to receive information from a camera and a microphone. The first computer includes an electronic processor. The electronic processor is configured to receive data including an image of a location and an audio of the location and generate a view of the location based on the received data. The electronic processor is configured to set a first sector of the view of the location in response to receiving a first user input, and set, a second sector of the view of the location in response to receiving a second user input. The electronic processor is configured to adjust a configuration of the first sector, in response to receiving a third user input, and adjust, a configuration of the second sector in response to receiving a fourth user input. The configuration of the second sector is different than the configuration of the first sector.

(5) Another example provides a method for controlling an audio analytics system. The method includes receiving, via an electronic processor, data including an image of a location from a camera and an audio of the location from a microphone, and generating, via the electronic processor and on a display, a view of the location based on the received data. The method includes setting, via the electronic processor and in response to receiving a first user input, a first sector of the view of the location, and setting, via the electronic processor and in response to receiving a second user input, a second sector of the view of the location. The method includes adjusting, via the electronic processor and in response to receiving a third user input, a configuration of the first sector, and adjusting, via the electronic processor and in response to receiving a fourth user input, a configuration of the second sector. The configuration of the second sector is different than the configuration of the first sector.

(6) Other features, aspects, and benefits of various embodiments will become apparent by consideration of the detailed description and accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a block diagram of an analytics system for configuring sectors of a camera according to one example.

(2) FIG. 2 is a block diagram of a first computer of the analytics system of FIG. 1 according to one example.

(3) FIG. 3 is a block diagram of a camera of the analytics system of FIG. 1 according to one example.

(4) FIG. 4 illustrates a camera view including multiple audio sectors according to one example.

(5) FIG. 5 illustrates a graphical user interface showing configurable audio sectors according to one example.

(6) FIG. 6 illustrates a flow chart of a method for configuring audio sectors of a camera according to one example.

(7) FIG. 7 illustrates a flow chart of a method for suppressing audio associated with a sector according to one example.

DETAILED DESCRIPTION

(8) One or more examples are described and illustrated in the following description and accompanying drawings. These examples are not limited to the specific details provided herein and may be modified in various ways. Other examples may exist that are not described herein. For example, a device or structure that is “configured” in a certain way is configured in at least that way but may also be configured in ways that are not listed.

(9) It should also be noted that a plurality of hardware and software based devices, as well as a plurality of different structural components may be utilized in various implementations. Aspects, features, and instances may include hardware, software, and electronic components or modules that, for purposes of discussion, may be illustrated and described as if the majority of the components were implemented solely in hardware. However, one of ordinary skill in the art, and based on a reading of this detailed description, would recognize that, in at least one instance, the aspects may be implemented in software (for example, stored on non-transitory computer-readable medium) executable by one or more processors. Thus, a plurality of hardware and software based devices, as well as a plurality of different structural components may be utilized to implement various aspects and examples. For example, “control units” and “controllers” described in the specification can include one or more electronic processors, one or more memory modules including a non-transitory computer-readable medium, one or more input/output interfaces, and various connections (for example, a system bus) connecting the components. It should be understood that although certain drawings illustrate hardware and software located within particular devices, these depictions are for illustrative purposes only. Unless the context clearly indicates otherwise, the illustrated components may be combined or divided into separate software, firmware and/or hardware. For example, instead of being located within and performed by a single electronic processor, logic and processing may be distributed among multiple electronic processors. Regardless of how they are combined or divided, hardware and software components may be located on the same computing device or may be distributed among different computing devices connected by one or more networks or other suitable communication links.

(10) Examples described herein may be implemented as non-transitory, computer-readable medium storing instructions executable by one or more electronic processors to perform the described functionality. As used in the present application, “non-transitory computer-readable medium” comprises all computer-readable media but does not include a transitory, propagating signal. Accordingly, non-transitory computer-readable medium may include, for example, a hard disk, a CD-ROM, an optical storage device, a magnetic storage device, ROM (Read Only Memory), RAM (Random Access Memory), register memory, a processor cache, other memory and storage devices, or combinations thereof.

(11) For ease of description, some or all of the example systems presented herein are illustrated with a single exemplar of each of its component parts. Some examples may not describe or illustrate all components of the systems. Other instances may include more or fewer of each of the illustrated components, may combine some components, or may include additional or alternative components.

(12) In addition, the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting. For example, the use of “including,” “containing,” “comprising,” “having,” and variations thereof herein is meant to encompass the items listed thereafter and equivalents thereof as well as additional items. The terms “connected” and “coupled” are used broadly and encompass both direct and indirect connecting and coupling. Further, “connected” and “coupled” are not restricted to physical or mechanical connections or couplings and can include electrical connections or couplings, whether direct or indirect. In addition, electronic communications and notifications may be performed using wired connections, wireless

connections, or a combination thereof and may be transmitted directly or through one or more intermediary devices over various types of networks, communication channels, and connections.

(13) Unless the context of their usage unambiguously indicates otherwise, the articles “a” and “an” should not be interpreted as meaning “one” or “only one.” Rather these articles should be interpreted as meaning “at least one” or “one or more.”

(14) Relational terms, for example, first and second, top and bottom, and the like may be used herein solely to distinguish one entity or action from another entity or action without necessarily requiring or implying any actual such relationship or order between such entities or actions.

(15) In some examples, method steps are conducted in an order that is different from the order described.

(16) FIG. 1 illustrates an analytics system **100** for monitoring a location. In the example illustrated, the analytics system **100** includes a plurality of cameras (such as a first camera **105A**, a second camera **105B**, and a third camera **105C**), a communications network **110**, a first computer **115**, and a database **125**. In the following description, when explaining how a single camera functions, a reference to camera **105** is used. It is not necessary that the cameras **105A** through **105C** be identical. Additionally, in some examples, the analytics system **100** includes more or fewer cameras than the number shown in FIG. 1.

(17) The cameras **105A** through **105C** communicate with the first computer **115** over the communications network **110**. The communications network **110** may be wired, wireless, or a combination thereof. All or parts of the communications network **110** may be implemented using various existing networks, for example, a cellular network, a Long Term Evolution (LTE) network, a 5G network, the Internet, network, a Bluetooth™ network, a wireless local area network (for example, Wi-Fi), a wireless a public switched telephone network, or other suitable networks. The communications network **110** may also include future developed networks. Also, in some embodiments, the communications network **110** includes a combination of networks. For example, the first camera **105A** may communicate over a LTE network, while the second camera **105B** may communicate over Wi-Fi.

(18) The first computer **115** is communicatively connected to the database **125**. In some examples the database **125** includes a memory (for example, a non-transitory, computer-readable medium) capable of storing images **130** and audio **135** received from the cameras **105A** through **105C** via the first computer **115**. In some examples, the first computer **115** includes a display and a user interface capable of receiving a plurality of user inputs.

(19) FIG. 2 is a block diagram of an example first computer **115**. The first computer **115** includes, among other things, a first electronic processor **205** (such as a programmable electronic microprocessor, microcontroller, or similar device), a first memory **210** (for example, a non-transitory, computer-readable medium), a first communication interface **215**, and a user interface **230**. In some instances, the first computer **115** also includes a plurality of electrical and electronic components that provide power, operational control, and protection to the components and modules within the first computer **115**. The first electronic processor **205**, the first memory **210**, the first communication interface **215**, and the user interface **230**, as well as the other various modules are connected by one or more control or data buses.

(20) The first electronic processor **205** obtains and provides information (for example, from the first memory **210** and the first communication interface **215**), and processes the information by executing one or more software instructions or modules capable of being stored, for example, in the first memory **210**). The software can include firmware, one or more applications, program data, filters, rules, one or more program modules, and other executable instructions. In some instances, the first electronic processor **205** executes instructions stored in the first memory **210** to perform the methods described herein. The first memory **210** can include one or more non-transitory computer-readable media and includes a program storage area and a data storage area. The program storage area and the data storage area can include combinations of different types of memory, for

example, read-only memory (“ROM”), random access memory (“RAM”), electrically erasable programmable read-only memory (“EEPROM”), flash memory, or other suitable digital memory devices.

(21) In some examples, the first memory **210** stores an audio analysis software **220** and a video analysis software **225**. The audio analysis software **220** is implemented by the first electronic processor **205** to analyze audio (for example, audio captured by a microphone included in the cameras **105A** through **105C** or an external device separate from the cameras **105A** through **105C**) and, among other things, determine types of sounds within the audio. Accordingly, the audio analysis software **220** (implemented by the first electronic processor **205**) receives data, including audio, from the cameras **105A** through **105C** and outputs information relating to the analyzed audio. For example, based on the received audio, the audio analysis software **220** may determine a gunshot, glass breaking, a break-in, or other noises present within the audio. The video analysis software **225** is implemented by the first electronic processor **205** to analyze images (for example, images **130** and/or video captured by the cameras **105A** through **105C** and stored in the database **125**) to, among other things, identify and determine features (such as objects of interest) within the images.

(22) In combination with the audio analysis software **220**, the video analysis software **225** may capture images associated with an event occurrence and communicate with the first electronic processor **205** to associate the images and the audio of the event occurrence. When the audio analysis software **220** determines an event occurrence based on the received audio, such as glass breaking, the audio analysis software **220** transmits an indication of the event occurrence and the event type (e.g., the output of the audio analysis software **220**) to the first electronic processor **205**. The first electronic processor **205** may associate the glass breaking event with corresponding video of the glass breaking. Accordingly, an operator may access an event and its corresponding video and audio.

(23) In some implementations, the audio analysis software **220** and the video analysis software **225** are machine-learning models pre-trained to identify particular features within provided data. Example machine learning techniques and include regression learning, classification learning, clustering analysis, density estimation, support vector machines, artificial neural networks, decision tree learning, and the like.

(24) As noted above, the first communication interface **215** allows the first computer **115** to communicate with other devices connected to the communications network **110**, such as the cameras **105A** through **105C**. The first electronic processor **205** is communicatively connected to the user interface **230**. The first electronic processor **205** receives user inputs from the user interface **230** and may configure settings of devices connected to the communications network **110** based on the user inputs, as described below in more detail. It should be understood that the functionalities of the first computer **115** may be spread over multiple devices and need not solely be performed within the first computer **115**.

(25) FIG. 3 is a block diagram of an example camera **105**. The camera **105** includes, among other things, an image sensor **305**, a microphone **310**, a second electronic processor **315**, a second memory **320**, and a second communication interface **325**. The image sensor **305** is configured to capture image and/or videos of a viewpoint of the camera **105**. In some examples, the image sensor **305** may be a lidar sensor, a radar sensor, or other sensor capable producing an image. The microphone **310** captures audio of the environment surrounding the camera **105**, including the location monitored by the image sensor **305**. In some situations, the microphone **310** is a micro electro-mechanical system (“MEMS”) sensor. In some examples, the microphone **310** is located within an external device but communicatively connected to the camera **105**.

(26) The second electronic processor **315** is communicatively connected to the image sensor **305**, the microphone, the second memory **320**, and the second communication interface **325**. It should be understood that the second electronic processor **315**, the second memory **320**, and the second

communication interface **325** include similar components to those described above with respect to the first electronic processor **205**, the first memory **210**, and the first communication interface **215**, respectively, and perform similar functions but need not be identical.

(27) While the audio analysis software **220** and the video analysis software **225** are described with respect to the first computer **115**, in some instances, the audio analysis software **220** and/or the video analysis software **225** are instead located within a camera **105**. For example, the camera **105** may apply the video analysis software **225** to captured video and apply the audio analysis software **220** to captured audio. The camera **105** then transmits the results of the video analysis and the audio analysis to the first computer **115**.

(28) In the examples described herein, individual sectors within the view of a camera may be configured based on a user's preferences. FIG. 4 illustrates a camera view **400** captured by the camera **105**. The camera **105** captures, with the image sensor **305**, the camera view **400** of an environment visible by the camera **105** (e.g., a location, a viewpoint). In the example shown in FIG. 4, the camera view **400** includes a first sector **405**, a second sector **410**, and a third sector **415** that, when combined, provide a 360 degree view surrounding the camera **105**. In the example provided, the first sector **405** provides a 90 degree view, the second sector **410** provides a 90 degree view, and the third sector **415** provides a 180 degree view. Accordingly, it is not necessary that the set angles of the first sector **405**, the second sector **410**, and the third sector **415** be identical. In some implementations, the camera view **400** includes more or fewer sectors than the number shown in FIG. 4. For example, two or four sectors may be provided. The camera **105** transmits the camera view **400**, via the communications network **110**, to the first computer **115**.

(29) In some instances, the camera **105** is in a fixed position to face a particular location. In other instances, the camera **105** is movable such that the camera **105** may view multiple locations. For example, a range of positions of a motor moving the camera **105** may define a sector. The first electronic processor **205** may determine which sector (or sectors) is being viewed based on the position of the motor.

(30) The sectors of FIG. 4 may be set using a graphical user interface. For example, FIG. 5 illustrates a graphical user interface **500** provided on the user interface **230** of the first computer **115**. The graphical user interface **500** includes a camera view **505**, a first sector **510**, a second sector **515**, and a plurality of interactive graphical elements **520**. The graphical user interface **500** allows for the individual configuration of each sector of the camera view **505** via the plurality of interactive graphical elements **520**. For example, the plurality of interactive graphical elements **520** includes radio buttons, check boxes, slider bars, and other elements that allow a user to alter settings of the sectors in the camera view **505**. In examples described herein, the first electronic processor **205** receives inputs to the graphical user interface **500**. Settings updated via the graphical user interface **500** are then provided to the second electronic processor **315** of the camera **105** via the communications network **110**.

(31) In the example of FIG. 5, the first sector **510** is selected for setting configuration. The plurality of interactive graphical elements **520** include enabling the detection of particular sounds, such as gunshots, glass breaking, screams, and an alarm via respective radio buttons. Slider bars are provided to adjust a sensitivity of detected sounds. In some implementations, the sensitivity indicates a sound threshold (for example, in decibels) that the sound should exceed for the second electronic processor to detect the sound event. In another implementation, the sensitivity indicates a confidence of the audio analysis software. When the confidence exceeds the set sensitivity value, the second electronic processor **315** detects the sound event. While not explicitly illustrated, the graphical user interface **500** may include additional interactive graphical elements **520**. For example, interactive graphical elements **520** may be provided for setting a size of each sector, setting priority values of the sectors, and scheduling configuration changes of the sectors, as described herein.

(32) In some examples, configuring the sensitivity of each sector includes configuring the

sensitivity of the microphone **310** (in decibels). For example, based on a user input via the plurality of interactive graphical elements **520**, the first electronic processor **205** configures the sensitivity of the microphone **310** detecting sound from the first sector **510**. In response to another user input via the plurality of interactive graphical elements **520**, the first electronic processor configures the sensitivity of the microphone **310** detecting sound from the second sector **515**.

(33) In other examples, the first electronic processor **205** configures the sensitivity of the audio analysis software **220** to particular sounds. For example, in response to a user input via the plurality of interactive graphical elements **520**, the first electronic processor **205** configures a confidence threshold of the audio analysis software **220** to detect the sound of a gunshot, the sound of glass breaking, the sound of an alarm, or the like. The sensitivity of each sound type may be set for each sector individually.

(34) In some instances, the first electronic processor **205** sets, based on inputs to the graphical user interface **500**, a priority value of the sectors of the camera view **400**. The priority value may be set based on a scale (e.g., a priority value on a scale of 0 through 10, a percentage between 0 and 100, or the like). The priority values indicate which sector should be prioritized given two simultaneous events in two different sectors. For example, if multiple events are detected in rapid succession from different sectors, the event from a higher priority sector will not be interrupted if it includes a “reaction” to the event (such as steering the camera, tracking a feature, or the like). An event in a lower priority sector may be interrupted by an event in a higher priority sector. Additionally, in some implementations, event types are also given priority values in addition to or in replacement of priority values for the sectors.

(35) As one example with respect to FIG. 5, the priority value of the first sector **510** is set to be greater than the priority value of the second sector **515**. In such an instance, the first sector **510** is deemed to be a greater priority than the second sector **515**. A first event occurrence located in the first sector **510** will not be interrupted by a second event occurrence located in the second sector **515**. In some instances, the first electronic processor **205** generates a notification identifying the first event occurrence on the graphical user interface **500**, but does not provide a notification identifying the second event occurrence.

(36) Additionally, in response to detected events, the camera **105** (or other devices within the analytics system **100**) may be controlled to perform an action. For example, the camera **105** may trigger an alarm, steer the camera **105** to the location of the detected event (i.e., if the camera **105** is a pan/tilt/zoom (“PTZ”) camera), or activate motion tracking in response to a detected event. A command indicating the action may be generated by the first computer **115** in response to the detected event, or may be generated and performed by the camera **105** itself in response to the detected event. Should the action be associated with a high-priority event or sector, the action is not interrupted given the detection of a second, lower-priority event.

(37) In some examples, the first electronic processor **205** sets a size of each sector of the camera view **400**. For example, in response to a first user input via the plurality of interactive graphical elements **520**, the first electronic processor **205** sets a size of the first sector **510** within the camera view **400** as a portion of the 360 degree view. In response to a second user input via the plurality of interactive graphical elements **520**, the first electronic processor **205** sets a size of the second sector **515** within the camera view **400** as a portion of the 360 degree view. In some examples, the size of each sector corresponds to a position of the camera **105**. For example, as previously described, the position of a motor used to actuate the camera **105** may be monitored to determine the location of the camera **105**, and therefore the viewpoint of the camera **105**. Sectors may be added or removed based on a user input via the plurality of interactive graphical elements **520**.

(38) In some examples, the first electronic processor **205** adjusts the configuration of each sector of the camera view **400** based on a time of day. For example, a user sets, via the plurality of interactive graphical elements **520**, a time at which a configuration for a sector of the camera view **400** changes. At the set time, the first electronic processor **205** may alter a size of a sector, alter a

priority of a sector or event, or the like. As one example, if the first sector **510** experiences a high level of noise at a particular time of day, the first electronic processor **205** decreases the sensitivity of the microphone **310** at that time of day. In another example, if the field of view of the second sector **515** includes a shooting range and gunshots are expected in the afternoon, the first electronic processor **205** decreases the sensitivity of the audio analysis software **220** detecting sound from the second sector **515** or decreases the sensitivity of the audio analysis software **220** to the sound of a gunshot. In another example, the first electronic processor **205** may disable the sensitivity to the sound of a gunshot for each sector of the camera view **400** during the Fourth of July such that an event occurrence will not be detected based on any nearby fireworks. Accordingly, the sector configurations may be dynamic based on times set by a user.

(39) FIG. **6** illustrates an example flowchart of method **600** for configuring the sectors of the camera view **400**. The method **600** is described as being executed by the first electronic processor **205**. However, in some examples, aspects of the method **600** are performed by another device. For example, the method **600** may be performed by the second electronic processor **315** of a camera **105** or the first electronic processor **205** in conjunction with the second electronic processor **315** of a camera **105**. Additionally, while a particular order is provided, in some examples, the steps of the method **600** may be performed in a different order.

(40) At step **605**, the first electronic processor **205** receives data including an image of a location and an audio of the location. For example, the first electronic processor **205** receives the camera view **505** and audio associated with the camera view **505** from the camera **105**. At step **610**, the first electronic processor **205** displays the image of the location. For example, as shown in FIG. **5**, the camera view **505** is displayed via the graphical user interface **500**.

(41) At step **615**, the first electronic processor **205** sets a first sector of view based on a first user input. For example, the first electronic processor **205** receives a first user input via the graphical user interface **500** indicative of an area of the first sector **510**. The first user input may, for example, indicate a portion of the 360 degree view of the camera **105**.

(42) At step **620**, the first electronic processor **205** sets a second sector of view based on a second user input. For example, the first electronic processor **205** receives a second user input via the graphical user interface **500** indicative of an area of the second sector **515**. The second user input may, for example, indicate a portion of the 360 degree view of the camera **105** that is not part of the first sector **510**. In some instances, the second sector **515** is set as a remaining portion of the 360 degree view of the camera **105**.

(43) At step **625**, the first electronic processor **205** adjusts a configuration of the first sector based on a third user input. For example, the first electronic processor **205** may set a priority value of the first sector **510**, a size of the first sector **510**, a sensitivity of the first sector **510**, or any command associated with the field of view of the camera view **505** to the first sector **510**, as previously described. At step **630**, the first electronic processor **205** adjusts a configuration of the second sector based on a fourth user input. For example, the first electronic processor **205** may set a priority value of the second sector **515**, a size of the second sector **515**, a sensitivity of the second sector **515**, or any command associated with the field of view of the camera view **505** to the second sector **515**, as previously described.

(44) It should be understood, that in other examples, the method **600** may continue to set and configure additional sectors of the camera view when the first electronic processor **205** receives additional user inputs. The method **600** may continue to set and configure as many additional sectors based on additional user inputs.

(45) In some implementations, prior to providing audio via an output device, the first electronic processor **205** suppresses audio that may be sensitive or confidential. FIG. **7** illustrates an example flowchart of method **700** for suppressing audio associated with the sectors of a camera view. It should be understood that the method **700** may be performed in conjunction with the method **600** as described above. Additionally, the method **700** is described as being executed by the first

electronic processor **205**. However, in some examples, aspects of the method **700** are performed by another device. For example, the method **700** may be performed by the second electronic processor **315** of a camera **105** or the first electronic processor **205** in conjunction with the second electronic processor **315** of a camera **105**. Additionally, while a particular order is provided, in some examples, the steps of the method **700** may be performed in a different order.

(46) At step **705**, the first electronic processor **205** receives audio from a microphone. For example, audio captured by the microphone **310** is transmitted from the camera **105** to the first computer **115**. At step **710**, the first electronic processor **205** determines whether a first sound is detected in the audio. For example, the first electronic processor **205** determines whether speech is present within the audio by implementing the audio analysis software **220**. For example, the first electronic processor **205** implements the audio analysis software **220** stored in the first memory **210** to determine whether a first sound is present within the audio. The first sound may include words, phrases, vocal frequency, or a vocal pitch. When the first electronic processor **205** determines that the first sound is present in the audio, the first electronic processor **205** proceeds to step **715**. When the first electronic processor **205** determines that the first sound is not present in the audio, the first electronic processor **205** proceeds to step **725**.

(47) At step **715**, the first electronic processor **205** determines the direction of the first sound. For example, the first electronic processor **205** may determine the direction of the first sound relative to the direction that the camera **105** is facing and transmit a command to the camera **105** to move the camera **105**. The camera **105** then moves to face the direction in which the first sound is detected. In other examples, the first electronic processor **205** may determine the direction of the first sound relative to a sector of the camera view **400**. In such an example, the direction of the first sound is determined to be within at least one of the sectors (e.g., the first sector **405**) of the camera view **400**. The first electronic processor **205** then transmits a command to the camera **105** to track motion within the camera viewpoint corresponding to the direction of the first sound. At step **720**, the first electronic processor **205** suppresses the first sound from the determined direction. For example, the first electronic processor **205** distorts the frequencies associated with the first sound or adds noise to the first sound to suppress the first sound. In some embodiments, the first electronic processor **205** selects and applies a filter to the first sound based on a configuration of the sector associated with the first sound.

(48) At step **725**, the first electronic processor **205** provides the audio to a user. For example, the first electronic processor **205** provides the audio via a speaker. In instances where the first sound is suppressed, the provided audio does not include the first sound, or includes a distorted version of the first sound.

(49) Thus, examples provide, among other things, a system and method for configuring audio sectors of a camera. Various features, advantages, and examples are set forth in the following claims.

Claims

1. An audio analytics system, the system comprising: a first computer configured to receive information from a camera and a microphone, the first computer including an electronic processor configured to: receive data including an image of a location and an audio of the location; generate a view of the location based on the received data; set, in response to receiving a first user input, a first sector of the view of the location; set, in response to receiving a second user input, a second sector of the view of the location; set, in response to receiving a third user input, a priority value of the first sector; set, in response to receiving a fourth user input, a priority value of the second sector, wherein the priority value of the second sector is different than the priority value of the first sector; detect a first event occurrence within the first sector based on data received from the microphone; detect a second event occurrence within the second sector based on data received from

the microphone; and generate a view of the first sector based on the priority value of the first sector and the first event occurrence.

2. The audio analytics system of claim 1, wherein the electronic processor is further configured to: disregard the second event occurrence based on the priority value of the second sector.

3. The audio analytics system of 1, wherein the electronic processor is further configured to: configure, in response to receiving a fifth user input, a sensitivity of the microphone detecting sound from the first sector; and configure, in response to receiving a sixth user input, a sensitivity of the microphone detecting sound from the second sector, the sensitivity of the second sector different than the sensitivity of the first sector.

4. The audio analytics system of claim 1, wherein the electronic processor is further configured to: set, in response to receiving the first user input, a size of the first sector within the view of the location; and set, in response to receiving the second user input, a size of the second sector within the view of the location, wherein the size of the second sector is different than the size of the first sector.

5. The audio analytics system of claim 1, wherein the electronic processor is further configured to: adjust a configuration of the first sector in response to a first time being satisfied; and adjust a configuration of the second sector in response to a second time being satisfied.

6. The audio analytics system of claim 5, wherein the first time is different from the second time.

7. The audio analytics system of claim 1, wherein the electronic processor is further configured to: detect a first sound associated with the first sector via the microphone; and in response to detecting the first sound, control the performance of an action with the camera, wherein the action is one selected from the group consisting of triggering an alarm, steering the camera to a location of the first sound, and activating motion tracking.

8. The audio analytics system of claim 1, wherein the electronic processor is further configured to: detect a first sound associated with the first sector via the microphone; and suppress, based on the configuration of the first sector, the first sound.

9. A method for controlling an audio analytics system, the method comprising: receiving, via an electronic processor, data including an image of a location from a camera and an audio of the location from a microphone; generating, via the electronic processor and on a display, a view of the location based on the received data; setting, via the electronic processor and in response to receiving a first user input, a first sector of the view of the location; setting, via the electronic processor and in response to receiving a second user input, a second sector of the view of the location; setting, via the electronic processor and in response to receiving a third user input, a priority value of the first sector; setting, via the electronic processor and in response to receiving a fourth user input, a priority value of the second sector, wherein the priority value of the second sector is different than the priority value of the first sector; detecting, via the electronic processor, a first event occurrence within the first sector based on data received from the microphone; detecting, via the electronic processor, a second event occurrence within the second sector based on data received from the microphone; and generating, via the electronic processor, a view of the first sector based on the priority value of the first sector and the first event occurrence.

10. The method of claim 9, further comprising: disregarding, via the electronic processor, the second event occurrence based on the priority value of the second sector.

11. The method of claim 9, further comprising: configuring, via the electronic processor and in response to receiving a fifth user input, a sensitivity of the microphone detecting sound from the first sector; and configuring, via the electronic processor and in response to receiving a sixth user input, a sensitivity of the microphone detecting sound from the second sector, the sensitivity of the second sector different than the sensitivity of the first sector.

12. The method of claim 9, further comprising: setting, via the electronic processor and in response to receiving the first user input, a size of the first sector within the view of the location; and setting, via the electronic processor and in response to receiving the second user input, a size of the second

sector within the view of the location, wherein the size of the second sector is different than the size of the first sector.

13. The method of claim 9, further comprising: adjusting, via the electronic processor, a configuration of the first sector in response to a first time being satisfied; and adjusting, via the electronic processor, a configuration of the second sector in response to a second time being satisfied.

14. The method of claim 13, wherein the first time is different than the second time.

15. The method of claim 9, further comprising: detecting, via the electronic processor, a first sound associated with the first sector via the microphone; and controlling, via the electronic processor, in response to detecting the first sound, the performance of an action with the camera, wherein the action is one selected from the group consisting of triggering an alarm, steering the camera to a location of the first sound, and activating motion tracking.

16. The method of claim 9, further comprising: detecting, via the electronic processor, a first sound associated with the first sector via the microphone; and suppress, via the electronic processor and based on the configuration of the first sector, the first sound.
